

Localization

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April 2025

The following is ATOCA Chapter 11, Exercise 5.

Lemma 0.1. Let R', R'' be rings. Let $R := R' \times R''$. Let $S = \{(1, 1), (1, 0)\}$. Then, R' is the localization of R at S .

Proof. We have a ring homomorphism from R onto R' given by the projection map with kernel R'' .

By ATOCA 11.10, $\ker(\phi_S) = \langle 0 \rangle^S = R''$.

Then, the induced map from $S^{-1}R$ to R' is injective. The map is also surjective since the diagram given in ATOCA 11.3 commutes. Thus, the two rings are isomorphic. $\square$

Let R be a ring and S be a multiplicative subset of R . Let J be an ideal of $S^{-1}R$. Let $\phi = \phi_S$. Let $I = \phi^{-1}(J)$.

Lemma 0.2. I is saturated.

Proof. Let $x \in I^S$. Then, there exists $s \in S$ such that $xs \in I$. Then, $\phi(xs) \in J$. But $\phi(s)$ is a unit in $S^{-1}R$, and thus we have that $\phi(x) = \phi(x)\phi(s)\phi(s^{-1}) = \phi(xs)\phi(s^{-1}) \in J$. Then, $x \in I$. $\square$

Lemma 0.3. $J = I(S^{-1}R)$.

Proof. Let $\frac{x}{s} \in J$. Then, $\frac{x}{1} \in J$. Then, $x \in I$. Then, $\frac{x}{1} \in I(S^{-1}R)$. Thus, $J \subseteq I(S^{-1}R)$.

Let $x \in I$. Then, $\frac{x}{1} \in J$. But then J contains $I(S^{-1}R)$ as J is closed under multiplication by $S^{-1}R$. $\square$

Now, let I be an ideal of R .

Lemma 0.4. $I(S^{-1}R)$ is a saturated ideal of $S^{-1}R$.

Proof. Let $x \in (I(S^{-1}R))^S$. Then, there exists some $s \in S$ such that $xs \in I(S^{-1}R)$. Since s is a unit in $S^{-1}R$, this implies that $x \in I(S^{-1}R)$. $\square$

Lemma 0.5. The preimage of $I(S^{-1}R)$ under ϕ is the saturation of I .

Proof. We show both inclusions.

Let $x \in I^S$. Then, there exists some $s \in S$ such that $xs \in I$. Then, $\phi(xs) \in J$, so $\phi(x) \in J$. Since $J = I(S^{-1}R)$, we have the desired result.

Let x be in the preimage of $I(S^{-1}R)$. Then, $\phi(x) \in I(S^{-1}R)$, so $\frac{x}{1} = \frac{y}{s}$ for some $y \in I$ and $s \in S$. Then, there is some $t \in S$ such that $xst = yt \in I$ since $y \in I$. Then, $s \in I^S$. $\square$

Let R be a ring and S be a multiplicative subset of R . Let P be a prime ideal of R .

Let $x \in P^S$. Then, there exists some $s \in S$ such that $xs \in P$. Then, either $x \in P$ or $s \in P$. Since $s \notin P$, $x \in P$. Thus, $P = P^S$.

Lemma 0.6. $P(S^{-1}R)$ is a prime ideal.

Proof. Since the preimage of $P(S^{-1}R)$ is $P^S = P$, $P(S^{-1}R)$ can't be a unit ideal.

Now, let $\frac{a}{s} \cdot \frac{b}{t} \in P(S^{-1}R)$. Then, $\frac{ab}{st} \in P(S^{-1}R)$ and thus $ab \in P$, and thus $a \in P$ or $b \in P$, so $\frac{a}{s} \in P(S^{-1}R)$ or $\frac{b}{t} \in P(S^{-1}R)$. $\square$

We thus gave a bijective correspondence that respects inclusions between saturated ideals of R and ideals of $S^{-1}R$. Notice that this bijection also respects prime ideals, which are always saturated.

The following is Exercise 11.27(c) from ATOCA.

Lemma 0.7. Let R be a ring, S be a multiplicative subset of R . Let I, J be ideals of R . Then, $(I^S J^S)^S = (IJ)^S$.

Proof. $\square$

The following is from ATOCA 12.11 and beyond.

Let R be a ring, S be a multiplicative subset of R and N be a submodule of M .

Let's first show that N^S is a submodule of M .

Let $x, y \in N^S$. Then, there exists $s, t \in S$ such that $sx + ty \in N$. Then, $(x + y)st = xst + yst \in N$ so $x + y \in N^S$. Moreover, for any $r \in R$, $rxs \in N$. Thus, N^S is a submodule. This gives us ATOCA 12.12(1a).

Notice that 12.12(1b) is an immediate consequence of the exactness of localization.

Lemma 0.8 (ATOCA 12.12(3b)). $S^{-1}N = S^{-1}N^S$.

Proof. Let $\frac{x}{s} \in S^{-1}N^S$.

Since $x \in N^S$, there exists some $t \in S$ such that $tx \in N$. Then, $\frac{tx}{s} \in N$ so $\frac{x}{s} = \frac{1}{t} \frac{tx}{s} \in N$. $\square$

Lemma 0.9 (ATOCA 12.17). Let R be a ring and S be a multiplicative subset of R . Let M be an R -module.

Then, $S^{-1}\text{Ann}(M) \subset \text{Ann}(S^{-1}M)$.

Proof. $\square$

Definition 1. Let R be a ring and M be an R -module. x is **nilpotent on M** if there exists some $n \geq 1$ such that $x^n M = 0$. In other words, $x \in \sqrt{\text{Ann}(M)}$.

We denote the set of such elements by $\text{nil}(M)$.

Notice that if $M = R$, we recover the usual definition of a nilpotent element.

Lemma 0.10 (ATOCA 12.23). Let R be a ring and I be an ideal of R . Then, $\text{nil}(R/I) = \sqrt{I}$.

Proposition 0.11. Let R be a ring and S be a multiplicatively closed subset of R . Let $Q \subsetneq M$ be R -modules.

Assume $S \cap \text{nil}(M/Q) \neq \emptyset$.

Then, $Q^S = M$ and $S^{-1}Q = S^{-1}M$.

Proof. Let $s \in S \cap \text{nil}(M/Q)$.

Then, there exists some $n \geq 1$ such that for every $m \in M$, $s^n m \in Q$. Since $s^n \in S$, this implies that $M = Q^S$.

Since $S^{-1}Q = S^{-1}Q^S$, we have the desired result. $\square$

Lemma 0.12 (ATOCA 12.40). Let R be a ring and S be a multiplicative subset of R . Let M be an R -module.

Then, $S^{-1}(\text{nil}(M)) \subseteq \text{nil}(S^{-1}M)$.

Proof. Let $\frac{x}{s} \in S^{-1}(\text{nil}(M))$ where $x \in \text{nil}(M)$ and $s \in S$.

Then, there exists some $n \geq 1$ such that x^n

Notice that $x \in$

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